Goole BigQuery exercise.

1.

```
SELECT*
FROM `mystic-sun-493417-r3.retail_sales.Data`
LIMIT 1000;

SELECT*
FROM `mystic-sun-493417-r3.retail_sales.Data`
WHERE Date BETWEEN '2023-01-01' AND '2023-12-31';
```

Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
191	2023-10-18	CUST191	Male	64	Beauty	1	25	25
204	2023-09-28	CUST204	Male	39	Beauty	1	25	25
230	2023-04-23	CUST230	Male	54	Beauty	1	25	25
232	2023-02-06	CUST232	Female	43	Beauty	1	25	25
309	2023-12-23	CUST309	Female	26	Beauty	1	25	25
310	2023-10-12	CUST310	Female	28	Beauty	1	25	25
363	2023-06-03	CUST363	Male	64	Beauty	1	25	25
371	2023-02-21	CUST371	Female	20	Beauty	1	25	25
397	2023-03-10	CUST397	Female	30	Beauty	1	25	25
454	2023-02-22	CUST454	Female	46	Beauty	1	25	25
512	2023-11-07	CUST512	Female	57	Beauty	1	25	25

2.

```
SELECT *
FROM `mystic-sun-493417-r3.retail_sales.Data`
WHERE `Total Amount` > (
    SELECT AVG(`Total Amount`)
    FROM `mystic-sun-493417-r3.retail_sales.Data`);
```

Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
21	2023-01-14	CUST021	Female	50	Beauty	1	500	500
28	2023-04-23	CUST028	Female	43	Beauty	1	500	500
128	2023-07-05	CUST128	Male	25	Beauty	1	500	500
220	2023-03-03	CUST220	Male	64	Beauty	1	500	500
238	2023-01-17	CUST238	Female	39	Beauty	1	500	500
364	2023-08-23	CUST364	Female	19	Beauty	1	500	500
408	2023-04-15	CUST408	Female	64	Beauty	1	500	500
537	2023-06-03	CUST537	Female	21	Beauty	1	500	500
539	2023-06-08	CUST539	Male	25	Beauty	1	500	500
541	2023-07-29	CUST541	Male	56	Beauty	1	500	500
760	2023-03-27	CUST760	Male	27	Beauty	1	500	500

3.

```
SELECT SUM(`Total Amount`) AS Total_Revenue
FROM `mystic-sun-493417-r3.retail_sales.Data`;
```

Total_Revenue
456000

4.

```
SELECT DISTINCT`Product Category` AS Product_Category
FROM `mystic-sun-493417-r3.retail_sales.Data`;
```

Product_Category
Beauty
Clothing
Electronics

5.

```
SELECT `Product Category` AS Product_Category,
SUM(`Quantity`) AS Total_Quantity
FROM `mystic-sun-493417-r3.retail_sales.Data`
GROUP BY `Product Category`;
```

Product_Category	Total_Quantity
Beauty	771
Clothing	894
Electronics	849

6.

```
SELECT `Customer ID`, `Age`,
 CASE
   WHEN Age<30 THEN 'Youth'
   WHEN Age BETWEEN 30 AND 59 THEN 'Adult'
   WHEN Age>=60 THEN 'Senior'
 END AS Age_Group
FROM  `mystic-sun-493417-r3.retail_sales.Data`;
```

Customer ID	Age	Age_Group
CUST191	64	Senior
CUST204	39	Adult
CUST230	54	Adult
CUST232	43	Adult
CUST309	26	Youth
CUST310	28	Youth
CUST363	64	Senior

7.

```
SELECT Gender,
  COUNT(*) AS High_Value_Transactions
FROM  `mystic-sun-493417-r3.retail_sales.Data`
WHERE `Total Amount` > 500
GROUP BY Gender;
```

Gender	High_Value_Tran...
Female	155
Male	144

8.

```
SELECT `Product Category` AS Product_Category,
SUM(`Total Amount`) AS Total_Revenue
FROM  `mystic-sun-493417-r3.retail_sales.Data`
GROUP BY `Product Category` HAVING(Total_Revenue>5000);
```

Product_Category	Total_Revenue
Beauty	143515
Clothing	155580
Electronics	156905

9.

```sql
SELECT `Transaction ID` AS Transaction_ID,
       `Price per Unit` AS Price_per_Unit,
 CASE
  WHEN `Price per Unit` <50 THEN 'Cheap'
  WHEN `Price per Unit` BETWEEN 50 AND 200 THEN 'Moderate'
  WHEN `Price per Unit` >200 THEN 'Expensive'
 END AS Unit_Cost_Category
 FROM `mystic-sun-493417-r3.retail_sales.Data`;
```

Transaction_ID	Price_per_Unit	Unit_Cost_Category
191	25	Cheap
204	25	Cheap
230	25	Cheap
232	25	Cheap
309	25	Cheap
310	25	Cheap
363	25	Cheap
371	25	Cheap
397	25	Cheap
454	25	Cheap
512	25	Cheap

10.

```sql
SELECT `Customer ID` AS Customer_ID,
       Age,
       `Total Amount` AS Total_Amount,
      CASE
        WHEN `Total Amount`  >1000 THEN 'High'
        ELSE 'Low'
      END AS Spending_Level
FROM `mystic-sun-493417-r3.retail_sales.Data`
WHERE Age>=40;
```

Customer_ID	Age	Total_Amount	Spending_Level
CUST191	64	25	Low
CUST230	54	25	Low
CUST232	43	25	Low
CUST363	64	25	Low
CUST454	46	25	Low
CUST512	57	25	Low
CUST791	51	25	Low
CUST825	46	25	Low
CUST855	54	25	Low

Kindly note that the attached screenshots represent the outcomes of the questions. For those with lengthy responses, the full answer could not be captured within the screenshot frame.